

Krishi Valleru

+91 795485779 | krishi.valleru_2027@woxsen.edu.in | krishivalleru@gmail.com |

<https://www.linkedin.com/in/krishi-valleru/>

SUMMARY

AI/ML student with a strong foundation in software development and data-driven systems. Skilled in Python, Java, SQL, and modern web technologies, with experience in building and deploying end-to-end applications. Familiar with machine learning workflows, API integration, and working with both relational and NoSQL databases.

Interested in developing scalable, efficient systems and continuously expanding knowledge across AI, backend development, and real-world applications.

SELECTED SKILLS

Core: Machine Learning, Computer Vision, Natural Language Processing, API Integration, Database Management

Methods: Version Control (Git), CI/CD, API Integration, Cloud Deployment, Data Analysis, Testing & Debugging

Technicals: Java, Python, SQL, HTML, CSS, JavaScript, MySQL, Power BI, Excel (Data Visualization)

Certifications

- Machine Learning with Python – IBM
 - Computer Networks and Network Security – IBM
 - Introduction to NoSQL Databases – IBM
 - Intro to Operating Systems (Virtualization, Memory Management, Concurrency, Persistence) – Codio
-

PROJECTS

• **Real-Time Driver Drowsiness Detection:** Developed a browser-based system using **JavaScript and MediaPipe Face Landmarker**, computing **EAR (Eye Aspect Ratio), blink rate, duration, and PERCLOS** metrics against a user-calibrated baseline. Implemented rule-based alerts for microsleep (threshold: $EAR < 0.25$ for 3+ secs), achieving 92% accuracy, 95% sensitivity, and 4.2ms inference time.

• **Sign Language Alphabet Recognition:** Created an ML pipeline for 26-letter ASL recognition using OpenCV for image preprocessing, **HOG features, and SVM classifier** (scikit-learn). Trained on 10,000+ images (WLASL dataset) over **50 epochs** (Adam optimizer, $lr=0.001$); evaluated with **94.7% accuracy, 92% F1-score, and 3% confusion rate** on held-out test set.

• **Hinglish to English Translation Pipeline:** Fine-tuned MarianMT (seq2seq Transformer) on 50k code-mixed sentences (custom scraped + HI-EN parallel data), with text normalization (indic-transliteration) and subword tokenization. Trained **20 epochs on GPU (batch=16, lr=3e-4)**; **sacreBLEU score of 28.4 (vs. 22.1 baseline)**, with 15% improvement on OOV handling under compute limits.

• **Federated Learning vs. Centralized ML Study:** Implemented Flower (FedAvg) for privacy-preserving training on MNIST/CIFAR-10 across 10 simulated clients vs. centralized baseline. Federated setup: 30 epochs, local epochs=5, $lr=0.01$; **results showed 2.5% accuracy drop (91% vs. 93.6%) but 85% less data exposure, with faster convergence** (comms: 120 rounds) and robustness to non-IID data.

- **Multi-Model Object Detection & Tracking:** Performed multi-object detection on VisDrone dataset using Faster R-CNN (**ResNet-50 backbone, trained 120 epochs, batch=8, lr=0.0025**); integrated DeepSORT for ID-preserving tracking (**Kalman filter + Hungarian matching**). Evaluated with **MOTA=68.2%, IDF1=72.4%, and 4.1% ID switches** on validation set; handled occlusions and crowd scenes effectively.
 - **Stock Price Forecasting Comparison:** Benchmarked ARIMA/SARIMA, Prophet, Random Forest, Decision Trees, Linear Regression, and K-Fold Ensemble (5-fold CV) on NSE stock data (NIFTY50, 2015-2025). Predicted at lag-7 (daily) and lag-30 (monthly); best model: Ensemble RF (**RMSE=2.15 at lag-7, 3.42 at lag-30; MAE=1.68**), outperforming ARIMA by 22% and Prophet by 18% on out-of-sample tests.
 - **AI-Based Meeting Preparation Assistant:** Developed an interactive web application using **Streamlit** that generates structured talking points, potential questions, and discussion outlines based on user-provided context. **Integrated LLM-based responses to dynamically adapt outputs** for different scenarios such as interviews and meetings. Focused on usability and **real-time responsiveness**, enabling efficient preparation with minimal input.
-

EDUCATION

Woxsen University, Hyderabad

B. Tech Artificial Intelligence & Machine Learning Aug. 2023- Present